Session M4 — Exponential Functions: Growth, Decay and Their Graphs

Duration: 45 minutes
Continues from: Session M1 (laws of exponents, rational exponents, same-base exponential equations, hidden quadratics) and the midterm review M3. Midterm score 94/100, so the pace is brisk and Tier 4 is a real challenge.
School unit: Unit 2, “Exponential and Logarithmic Functions” (Georgia standard AA.FGR.3: inverse functions, exponential and logarithmic graphs, logarithm properties, modeling). This session is the first of the unit.
Foundation: R.D. Sharma *Exponents of Real Numbers* (laws of exponents), already covered in M1.
School textbook: Big Ideas *Advanced Algebra with CalcChat and CalcView* (Larson & Boswell), chapter “Exponential and Logarithmic Functions”: the lessons on exponential growth and decay functions, and the natural base e.
Why now: Unit 2 starts when school resumes. Every later idea in the unit (inverse functions, logarithms, solving exponential equations) is built on the functions introduced today.

Lesson Plan

Objectives. By the end of this session the student should be able to:

1. Tell a linear pattern (constant difference) from an exponential one (constant ratio).
2. Define $f(x) = a \cdot b^x$, explain the restrictions on b, and state its key features.
3. Write growth and decay models $y = a(1 + r)^t$ and $y = a(1 - r)^t$, including half-life and doubling time.
4. Derive and use the compound-interest formula, and explain where e comes from.
5. Graph $y = a \cdot b^{x-h} + k$ from its asymptote and anchor points.
6. Write an exponential function through two points, and solve equations by a common base.

Time	Segment
0–4 min	Warm-up: 4 exponent questions
4–14 min	Part 1: linear vs exponential, the definition and key features
14–27 min	Part 2: growth and decay models, compound interest and e
27–35 min	Part 3: graphs and transformations
35–40 min	Part 4: writing from two points, and solving
40–45 min	Exit ticket, assign homework

Pacing note. Part 2 carries the most weight on school tests (word problems). If time is short, cut Part 4 to the first two-point example and set the rest as homework Q11–Q13.

Warm-up (4 min)

1. Simplify $\left(2x^3\right)^2 \cdot x^{-4}$. → $4x^6 \cdot x^{-4} = 4x^2$
2. Evaluate $32^{3/5}$. → $\left(\sqrt[5]{32}\right)^3 = 2^3 = 8$
3. Evaluate $5^0 + 4^{-1/2}$. → $1 + \frac{1}{\sqrt{4}} = 1 + \frac{1}{2} = \frac{3}{2}$
4. Solve $3^{x+2} = 81$. → $81 = 3^4$, so $x + 2 = 4$ and $x = 2$

Bridge to today: so far the exponent was a fixed number or the unknown in an equation. Today the exponent becomes the **input** of a function: $f(x) = 2^x$.

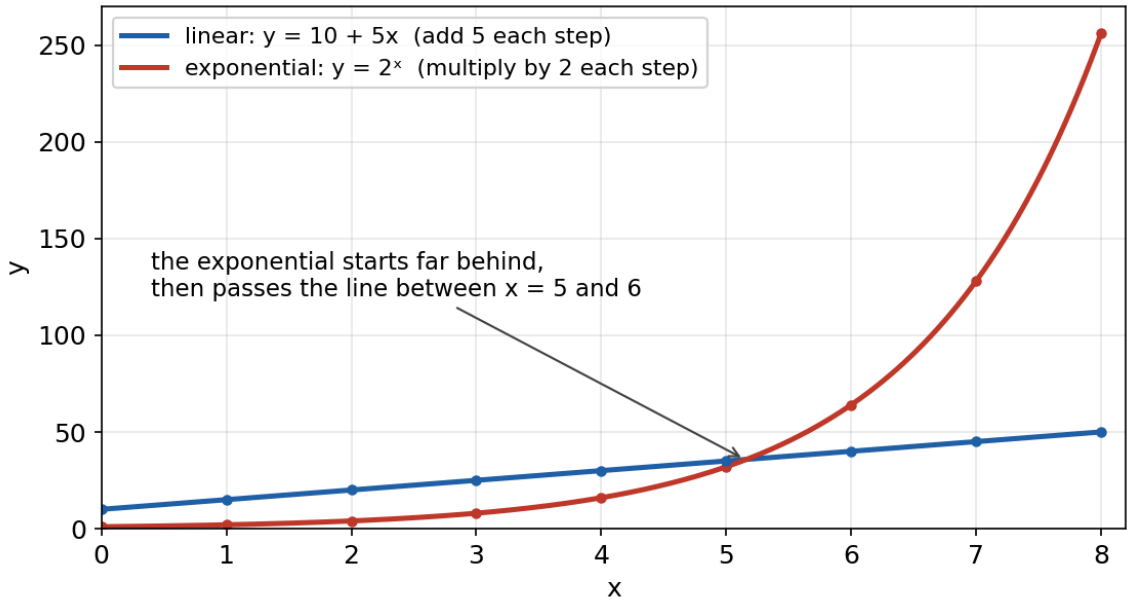
Part 1 — Linear vs Exponential (10 min)

Constant difference vs constant ratio

x	0	1	2	3	4
$f(x) = 3x + 2$	2	5	8	11	14
$g(x) = 2 \cdot 3^x$	2	6	18	54	162

- f **adds 3** each time x goes up by 1: a constant **difference**. That is a linear function.
- g **multiplies by 3** each time x goes up by 1: a constant **ratio**. That is an exponential function.

Test for a table with equally spaced x-values: subtract consecutive y-values (all equal → linear); divide them (all equal → exponential).



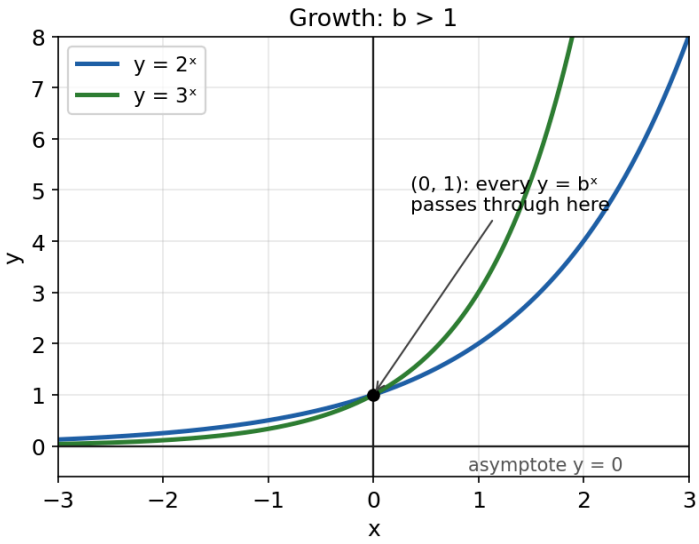
Definition

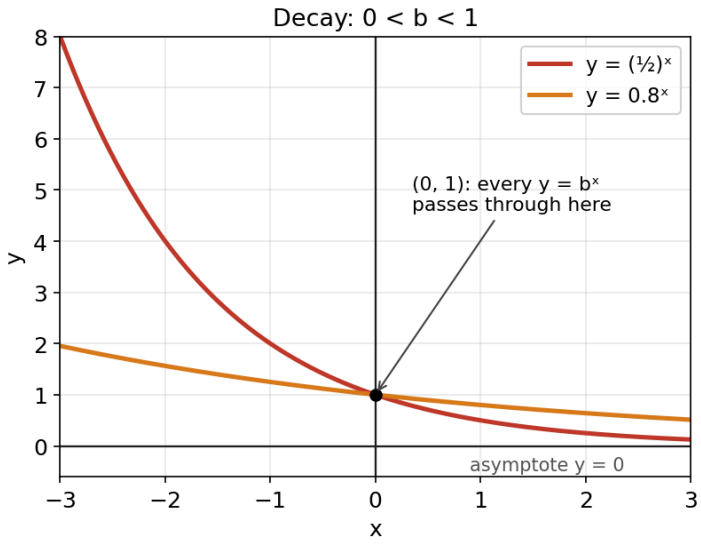
> An **exponential function** has the form
>
> $$f(x) = a \cdot b^x, \qquad a \neq 0,\ b > 0,\ b \neq 1$$
>
> a is the **initial value** ($f(0) = a \cdot b^0 = a$) and b is the **base**, the constant ratio (the growth or decay factor).

Why these restrictions on b:

- **b > 0:** with a negative base, $(-2)^{1/2} = \sqrt{-2}$ is not a real number, so the function would have gaps; and its values would flip sign at every whole number: $(-2)^1 = -2$, $(-2)^2 = 4$.
- **b ≠ 1:** $1^x = 1$ for every x, which is a constant (linear) function, not an exponential one.
- **a ≠ 0:** $0 \cdot b^x = 0$ for every x.

Growth and decay





Key feature (for $a > 0$)	Value	Reason
y-intercept	$(0, a)$	$b^0 = 1$
Horizontal asymptote	$y = 0$	b^x gets closer and closer to 0 but never reaches it
Domain	all real numbers	any real power of a positive number is defined
Range	$y > 0$	a positive number to any power is positive
Growth or decay	growth if $b > 1$ (increasing); decay if $0 < b < 1$ (decreasing)	multiplying by more than 1 increases, by less than 1 decreases
End behavior (growth)	$y \to \infty$ as $x \to \infty$; $y \to 0$ as $x \to -\infty$	
End behavior (decay)	$y \to 0$ as $x \to \infty$; $y \to \infty$ as $x \to -\infty$	

If $a < 0$, the graph is reflected across the x-axis: the range becomes $y < 0$.

Part 2 — Growth and Decay Models (13 min)

Derivation: the percent model

A quantity starts at a and grows by a fixed percent r (as a decimal) each period.

- After 1 period: $a + ar = a\,(\,1 + r\,)$.
- After 2 periods: $a\,(\,1 + r\,)\cdot(\,1 + r\,) = a\,(\,1 + r\,)^{\,2}$.
- After t periods: multiply by $(\,1 + r\,)$ t times:

$$\text{growth: } y = a\,(\,1 + r\,)^{\,t} \qquad\qquad \text{decay: } y = a\,(\,1 - r\,)^{\,t}$$

The **growth factor** is $b = 1 + r$; the **decay factor** is $b = 1 - r$. Read the rate back from the base: $b = 1.07$ means growth of 7%; $b = 0.85$ means decay of 15% (because $1 - 0.85 = 0.15$).

Worked Example 1: population growth

A town of 12,000 people grows 3% per year. Write a model and predict the population after 10 years.

- $a = 12\,000$, $r = 0.03$, so $P(\,t\,) = 12\,000\,(\,1.03\,)^{\,t}$.
- $P(\,10\,) = 12\,000\,(\,1.03\,)^{\,10} = 12\,000 \times 1.3439 = 16\,127$ people.

Worked Example 2: depreciation

A car bought for $28,000 loses 15% of its value each year. Find its value after 5 years.

- Decay factor $b = 1 - 0.15 = 0.85$: $V(\,t\,) = 28\,000\,(\,0.85\,)^{\,t}$.
- $V(\,5\,) = 28\,000\,(\,0.85\,)^{\,5} = 28\,000 \times 0.44371$ = $12,423.75.

Half-life and doubling time

If a quantity halves every h time units, then after t units it has halved t/h times:

$$y = a\left(\tfrac{1}{2}\right)^{t/h} \qquad\qquad \text{and if it doubles every } d\text{:}\quad y = a\cdot 2^{t/d}$$

Worked Example 3. A 400 mg dose of a medicine has a half-life of 6 hours.

- After 24 hours (4 half-lives): $400\left(\tfrac{1}{2}\right)^{24/6} = 400 \times \tfrac{1}{16} = 25$ mg.
- After 9 hours: $400\left(\tfrac{1}{2}\right)^{1.5} = 141$ mg. The exponent doesn't have to be a whole number.

Worked Example 4. A bacteria culture of 500 cells doubles every 20 minutes. After 2 hours (120 min) it has doubled $120/20 = 6$ times: $500 \cdot 2^{6} = 32\,000$ cells.

Derivation: compound interest

A principal P earns an annual rate r, compounded n times a year. Each compounding period pays a rate of r/n, so the money is multiplied by $(\,1 + r/n\,)$ each period. In t years there are nt periods:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Worked Example 5. $5,000 is invested at 4% for 10 years.

Compounding	n	Amount after 10 years
annually	1	$5000\,(\,1.04\,)^{\,10}$ = $7,401.22

Compounding	n	Amount after 10 years
monthly	12	$5000\,(\,1+0.04/12\,)^{\,120}$ = $7,454.16
daily	365	$5000\,(\,1+0.04/365\,)^{\,3650}$ = $7,458.96
continuously	—	$5000e^{0.4}$ = $7,459.12

More frequent compounding pays more, but the gain shrinks: it approaches a limit.

Where the number e comes from

Invest $1 at 100% interest for one year, compounded n times: $A = \left(1+\frac{1}{n}\right)^{n}$.

n	$\left(1+\frac{1}{n}\right)^{n}$
1	2
2	2.25
4	2.4414
12	2.6130
365	2.7146
10,000	2.7181
1,000,000	2.7182805

The values level off at **e = 2.718281828…**, an irrational number like π. It is the most a dollar can grow in a year at 100% interest.

Continuous compounding. Write $m = n/r$, so $n = mr$ and $\frac{r}{n} = \frac{1}{m}$:

$$\left(1+\frac{r}{n}\right)^{nt} = \left(1+\frac{1}{m}\right)^{mrt} = \left[\left(1+\frac{1}{m}\right)^{m}\right]^{rt}$$

As n grows without limit, so does m, and the bracket approaches e:

$$A = Pe^{rt}$$

Part 3 — Graphs and Transformations (8 min)

$$y = a \cdot b^{x-h} + k$$

Parameter	Effect on the graph of $y = b^x$
h	shifts right by h (left if h is negative)
k	shifts up by k; the **asymptote moves to y = k**
|a| > 1 or < 1	vertical stretch or shrink
a < 0	reflection across the asymptote (over the x-axis before the shift)

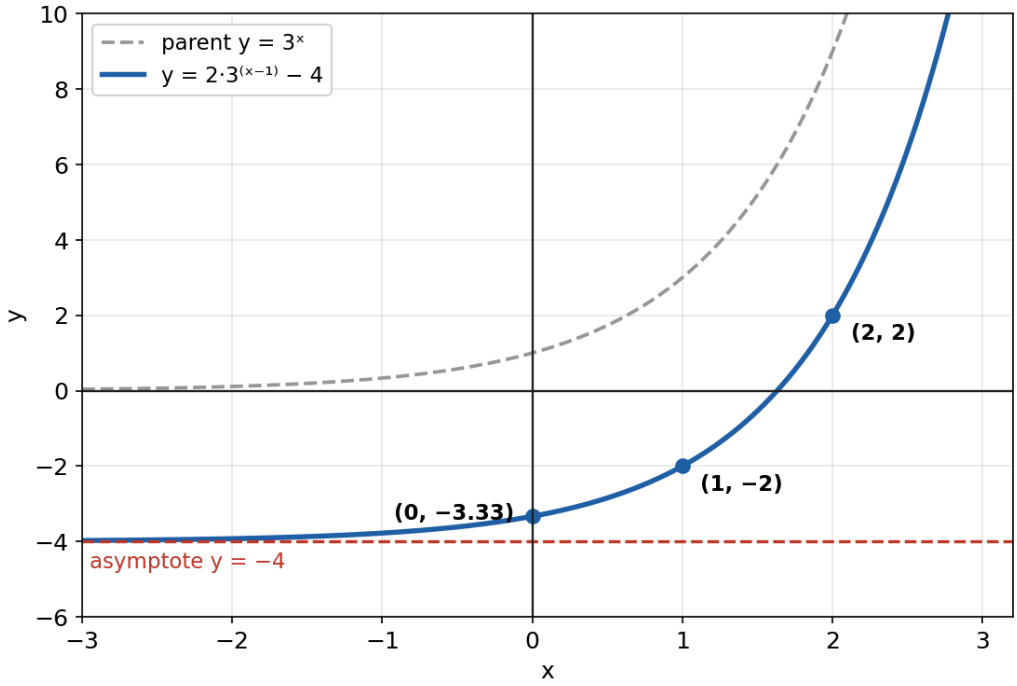
Sketching method:

1. Draw the asymptote $y = k$ as a dashed line.
2. Plot an anchor point at $x = h$: there $b^0 = 1$, so $y = a + k$.
3. Plot one or two more points (x = h + 1 and x = h − 1, or the y-intercept).

4. Draw the curve approaching the asymptote on one side.

Worked Example 6. Graph $y = 2 \cdot 3^{x-1} - 4$.

- Asymptote $y = -4$; range $y > -4$; domain all real numbers.
- $x = 1$: $y = 2(1) - 4 = -2$. $x = 2$: $y = 2(3) - 4 = 2$. $x = 0$: $y = 2\left(\frac{1}{3}\right) - 4 = -\frac{10}{3} \approx -3.33$ (the y-intercept).
- The graph crosses the x-axis between $x = 1$ and $x = 2$, where $3^{x-1} = 2$. To find that x exactly, you need a logarithm (next session): $x = 1 + \log_3 2 \approx 1.63$.



Part 4 — Writing from Two Points, and Solving (5 min)

Worked Example 7: two points

Find $y = a \cdot b^x$ through $(1, 6)$ and $(3, 54)$.

- Substitute both: $6 = ab$ and $54 = ab^3$.
- **Divide** the second by the first to cancel a: $\frac{ab^3}{ab} = \frac{54}{6}$, so $b^2 = 9$ and $b = 3$ (b must be positive).
- Then $a = 6/b = 2$. **Answer:** $y = 2 \cdot 3^x$. Check: $2 \cdot 3^3 = 54$ ✓

Solving

- **Same base:** $8^x = 32$: $2^{3x} = 2^5$, so $x = \frac{5}{3}$.
- **Bases don't match:** when does the town in Example 1 double? Solve $1.03^t = 2$. A table gives $1.03^{23} = 1.974$ and $1.03^{24} = 2.033$, so it doubles **between 23 and 24 years**. The exact answer needs the **logarithm**, the inverse of an exponential function, which is next session: $t = \frac{\log 2}{\log 1.03} \approx 23.4$ years.

Exit Ticket (3 min)

1. Is $y = 5\,(\,0.92\,)^{\,t}$ growth or decay? What is the percent rate?
2. $2,000 is invested at 6% compounded quarterly. Find the amount after 3 years.
3. Give the asymptote and the range of $y = -3 \cdot 2^{x} + 5$.

Practice Problems

Calculator allowed. Round money to the cent. Do Q4, Q9 and Q12 in class if time allows; the rest are homework.

Tier 1 — Recognize and describe

1. Growth or decay? Give the percent rate. (a) $y = 300\,(\,1.07\,)^{\,t}$ (b) $y = 50\,(\,0.6\,)^{\,t}$ (c) $y = 2\,(\,1.5\,)^{\,x}$ (d) $y = 900\left(\tfrac{1}{3}\right)^{x}$
2. For $f\,(\,x\,) = 4\left(\tfrac{1}{2}\right)^{x}$, give the y-intercept, asymptote, domain, range, whether it is increasing or decreasing, and $f\,(\,-2\,)$.
3. Linear, exponential or neither? (a) y: 5, 10, 20, 40 (b) y: 3, 7, 11, 15 (c) y: 1, 2, 4, 7 (x = 0, 1, 2, 3 in each).

Tier 2 — Models

4. A town of 8,500 people shrinks 2% per year. Write the model and find the population after 6 years.
5. A $1,200 laptop loses 25% of its value each year. Find its value after 3 years.
6. $3,000 is invested at 5% for 8 years. Find the amount if interest is compounded (a) annually, (b) monthly, (c) continuously.
7. Caffeine has a half-life of about 5 hours. You drink 200 mg at 8 a.m. How much remains at (a) 6 p.m., (b) 11 p.m.?
8. A video has 1,500 views, and its views double every 3 days. Write a model and find the views after 15 days.

Tier 3 — Graphs, two points, solving

9. Graph $y = 3 \cdot 2^{x+1} - 6$. Give the asymptote, three points, the domain, the range, and both intercepts.
10. Describe how to get $y = -2^{x-3} + 1$ from $y = 2^{x}$. Give its asymptote and range.
11. Write $y = ab^{x}$ through $(\,0, 7\,)$ and $(\,2, 63\,)$.
12. Write $y = ab^{x}$ through $(\,1, 10\,)$ and $(\,4, 80\,)$.
13. Solve: (a) $4^{x+1} = 8^{x}$ (b) $\left(\tfrac{1}{9}\right)^{x} = 27$ (c) $5^{2x-1} = 125$

Tier 4 — Challenge

14. **Two banks.** Bank A pays 6% compounded annually; Bank B pays 5.9% compounded monthly. Which is better for $10,000 over 10 years, and by how much? Explain why the lower rate wins.
15. **Doubling time.** A town grows 4% per year. Use a table to find between which two whole years it doubles. Compare with the “rule of 72” estimate, 72 ÷ 4.

16. **Cooling coffee.** Coffee at 190°F sits in a 70°F room. Its temperature after t minutes is $T(t) = 70 + 120(0.95)^t$. (a) What does the asymptote mean here? (b) Find T(10). (c) After how many whole minutes is the coffee first below 120°F?

Answer Key — Full Step-by-Step Solutions

Exit ticket

1. $b = 0.92 < 1$, so **decay**. $1 - 0.92 = 0.08$: **8% per period**.
2. $n = 4$, so the quarterly rate is $0.06/4 = 0.015$ over $4 \times 3 = 12$ quarters: $A = 2000(1.015)^{12}$ = **$2,391.24**.
3. k = 5, so the asymptote is **y = 5**. a = −3 < 0 reflects the graph below the asymptote, so the range is **y < 5**.

1. Growth or decay

- **(a)** $b = 1.07 > 1$: growth, $r = 0.07$, **7%**.
- **(b)** $b = 0.6$: decay, $1 - 0.6 = 0.4$, **40%**.
- **(c)** $b = 1.5$: growth, **50%**.
- **(d)** $b = \frac{1}{3}$: decay, $1 - \frac{1}{3} = \frac{2}{3}$, **about 66.7%**.

2. $f(x) = 4\left(\frac{1}{2}\right)^x$

- y-intercept $(0, 4)$; asymptote $y = 0$; domain all real numbers; range $y > 0$; **decreasing** (b < 1).
- $f(-2) = 4\left(\frac{1}{2}\right)^{-2} = 4 \times 2^2 = 16$.

3. (a) Ratios $10/5 = 20/10 = 40/20 = 2$: **exponential**. (b) Differences all 4: **linear**. (c) Differences 1, 2, 3 and ratios 2, 2, 1.75: **neither**.

4. $P(t) = 8500(0.98)^t$, and $P(6) = 8500(0.98)^6 = 8500 \times 0.88584$ = **about 7,530 people**.

5. $V(3) = 1200(0.75)^3 = 1200 \times 0.421875$ = **$506.25**.

6.

- **(a)** $3000(1.05)^8$ = **$4,432.37**
- **(b)** $3000\left(1 + \frac{0.05}{12}\right)^{96}$ = **$4,471.76**
- **(c)** $3000e^{0.05 \times 8} = 3000e^{0.4}$ = **$4,475.47**

7. $A(t) = 200\left(\frac{1}{2}\right)^{t/5}$.

- **(a)** 6 p.m. is 10 hours later: $200\left(\frac{1}{2}\right)^2$ = **50 mg**.
- **(b)** 11 p.m. is 15 hours later: $200\left(\frac{1}{2}\right)^3$ = **25 mg**.

8. $V(t) = 1500 \cdot 2^{t/3}$. After 15 days: $1500 \cdot 2^5 = 1500 \times 32$ = **48,000 views**.

9. $y = 3 \cdot 2^{x+1} - 6$

- Asymptote $y = -6$ (k = −6).

- Anchor at $x = -1$ (where the exponent is 0): $y = 3 - 6 = -3$. Also $x = 0$: $y = 3 \cdot 2 - 6 = 0$; $x = 1$: $y = 3 \cdot 4 - 6 = 6$.
- Domain all real numbers; range $y > -6$.
- The graph passes through the **origin**: the y-intercept is $(0, 0)$ and the x-intercept is $x = 0$.

10. Shift $y = 2^x$ **right 3**, **reflect it over the x-axis** (the minus sign is outside the power), then shift it **up 1**. Asymptote $y = 1$; because of the reflection the graph lies below it, so the range is $y < 1$.

11. $(0, 7)$ gives $a = 7$ directly ($b^0 = 1$). Then $63 = 7b^2$, so $b^2 = 9$ and $b = 3$. $y = 7 \cdot 3^x$

12. $10 = ab$ and $80 = ab^4$. Divide: $b^3 = 8$, so $b = 2$. Then $a = 10/2 = 5$. $y = 5 \cdot 2^x$. Check: $5 \cdot 2^4 = 80$ ✓

13.

- **(a)** $2^{2(x+1)} = 2^{3x}$, so $2x + 2 = 3x$ and **x = 2**. Check: $4^3 = 64 = 8^2$ ✓
- **(b)** $3^{-2x} = 3^3$, so $-2x = 3$ and $x = -\frac{3}{2}$.
- **(c)** $125 = 5^3$, so $2x - 1 = 3$ and **x = 2**.

14.

- Bank A: $10\,000(1.06)^{10}$ = $17,908.48.
- Bank B: $10\,000\left(1 + \frac{0.059}{12}\right)^{120}$ = $18,013.82.
- **Bank B is better by $105.34.** Monthly compounding gives an effective yearly rate of $\left(1 + \frac{0.059}{12}\right)^{12} - 1 = 0.0606$, i.e. 6.06%, which beats 6% compounded once a year.

15. $1.04^{17} = 1.948$ and $1.04^{18} = 2.026$, so the town doubles **between 17 and 18 years**. The rule of 72 gives $72/4 = 18$ years, a good estimate. (Exact, with logarithms: 17.7 years.)

16.

- **(a)** As t grows, $120(0.95)^t \to 0$ and $T \to 70$: the coffee approaches **room temperature, 70°F**, but never goes below it.
- **(b)** $T(10) = 70 + 120(0.95)^{10} = 70 + 71.85$ = **141.85°F**.
- **(c)** Need $120(0.95)^t < 50$. Check whole minutes: $T(17) = 70 + 50.17 = 120.17$°F (not yet below 120); $T(18) = 70 + 47.67 = 117.67$°F. **18 minutes.**

Most common mistakes

1. **Using the rate instead of the factor:** decay of 15% means $b = 0.85$, not $b = 0.15$ or $b = 1.15$.
2. **Applying the coefficient before the power:** $2 \cdot 3^x$ is not 6^x. The exponent applies only to the 3.
3. **Forgetting to divide the rate and multiply the time** in compound interest: monthly at 4% for 10 years is $(1 + 0.04/12)^{120}$.
4. **Moving the asymptote the wrong way:** only k moves it; h never does.
5. **Taking $b = -3$ from $b^2 = 9$:** the base of an exponential function must be positive.
6. **Writing the half-life exponent upside down:** it is t/h (elapsed time ÷ half-life), not h/t.

Notes for next session

- **M5 — Inverse functions and logarithms:** inverses as “undo” functions (swap x and y, reflect over y = x), defining $\log_b x$ as the inverse of b^x, converting between exponential and logarithmic form, common log and natural log (ln, base e), and graphs of $y = \log_b x$.
- **M6 — Properties of logarithms:** product, quotient and power rules (each derived from a law of exponents), change of base.
- **M7 — Solving exponential and logarithmic equations and modeling:** today’s “between 23 and 24 years” questions answered exactly; extraneous solutions of log equations.
- **Gifted extensions to watch for:** logistic growth, systems with exponentials, a change-of-base proof.